St. Francis Institute of Technology
Department of Computer Engineering

Criminal Code Summarization and Outcome Prediction Using InLegalBERT and Bharatiya Nyaya Sanhita Mapping

Group Members

Name	Roll No.
Samit Fernandes	03
Jaden Franco	04
Ralph Pereira	19

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Introduction

- The Bharatiya Nyaya Sanhita (BNS) Act introduces a modernized legal framework, replacing the long-standing Indian Penal Code (IPC), which has faced challenges due to outdated language and complex classifications.
- The BNS Act simplifies these classifications, aiming to enhance accessibility, efficiency, and relevance in criminal law processes.
- The transition to the BNS framework presents challenges for legal professionals who rely on IPC-based tools, as existing systems are not compatible with the new framework.

Introduction

- There is a need for BNS-specific tools and datasets to assist with tasks such as legal document summarization, section mapping, and case outcome prediction.
- This project aims to bridge the gap by creating a structured dataset and a rule-based mapping strategy to align IPC sections with their BNS equivalents, improving legal document processing under the new BNS framework.

Existing System

- Existing systems are designed for IPC-based analysis and lack the structural adjustments needed for the BNS Act, resulting in a lack of BNS-compatible datasets.
- There is no systematic mapping of IPC sections to their BNS equivalents, making it difficult to interpret or apply historical data within the new framework.
- Current legal document summarization models are tailored to IPC-based cases, failing to align with the simplified classifications of BNS.

Literature Review

- Paul, et al. [1] retrained LegalBERT and CaseLawBERT on Indian legal data, finding that InLegalBERT showed significant improvements in legal document summarization, especially in understanding legal terminology and context.
- S. Sharma et al. [2] explores using InLegalBERT for legal document summarization. Evaluated against Legal Pegasus, T5 base, BART, and BERT using ROUGE-L F1 scores, InLegalBERT achieves the best performance for Indian legal texts.

Literature Review

- Rani, U. and Bidhan, K. [3] provides a comparative analysis of various extractive summarization techniques, including TextRank, TF-IDF, and LDA, evaluating their effectiveness in condensing textual content, especially in contexts requiring concise and relevant summaries .

Problem Definition

The shift from IPC to BNS creates challenges in legal document analysis due to the lack of a BNS-specific dataset and tools. Existing systems, designed for IPC, cannot efficiently handle the new BNS structure, hindering tasks like summarization, section mapping, and outcome prediction. This highlights the need for tailored tools and datasets to support legal analysis under the BNS framework.

Proposed Solution

1. Create synthetic BNS dataset mapping IPC to BNS
2. Use InLegalBERT for generating embeddings for extractive summarization
3. Fine-tune BART for abstractive summarization
4. Present case findings in detail
5. Rule-based IPC to BNS mapping

Proposed Solution

1. **Create synthetic BNS dataset mapping IPC to BNS**
    - Since no official BNS dataset exists, a synthetic dataset was created.
    - IPC sections were manually mapped to their corresponding BNS sections.
    - This dataset serves as the foundation for training and testing the system on BNS cases.
    - Ensures structured analysis and continuity between IPC and BNS cases.

Proposed Solution

2. **Use InLegalBERT for generating embeddings for extractive summarization**
   - InLegalBERT, a legal-domain pre-trained model, is used for sentence embedding.
   - Each sentence from the legal document is transformed into a dense vector.
   - Sentence similarity is computed using embeddings to select key sentences.
   - Helps extract the most relevant parts of the document while preserving legal meaning.

Proposed Solution

3. **Fine-tune BART for abstractive summarization**
   - A fine-tuned BART model (facebook/bart-large) is used for summarization.
   - Extracted sentences are further processed to generate human-like, rephrased summaries.
   - BART handles complex legal terminologies and produces coherent, fluent summaries.
   - This improves readability without losing important legal facts.

Proposed Solution

4. **Present case findings in detail**
    - After summarization, detailed case analysis is performed.
    - Important aspects like case background, evidence, legal sections, and outcomes are extracted.
    - Summarization is complemented with IPC and BNS code references.
    - Helps legal professionals understand the case in a concise but informative manner.

Proposed Solution

5. **Rule-based IPC to BNS mapping**
   - A rule-based system identifies IPC sections in the text using regex and NER (Named Entity Recognition).
   - It then maps them automatically to the corresponding BNS sections.
   - The mapping ensures that legal documents can be interpreted accurately under the new BNS framework.
   - This strategy bridges the gap between historical IPC cases and the new legal structure.

Flowchart

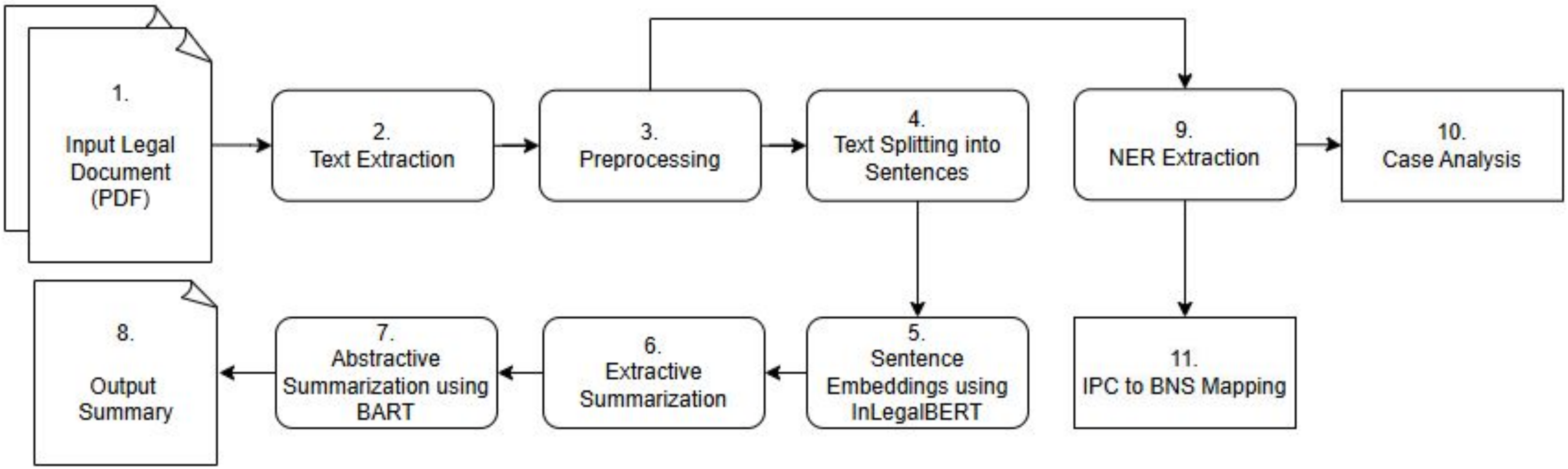


fig. (1) Flowchart for document summarization and sections mapping

Flowchart

1. **Input Legal Document**
2. **Text Extraction from the document**
3. **Preprocessing (cleaning, formatting)**
4. **Text Splitting into sentences**
5. **Sentence Embeddings using InLegalBERT**
6. **Extractive Summarization**
7. **Abstractive Summarization using BART**
8. **Output Final Summary**

Flowchart

9. **NER Extraction (find legal entities like dates, sections)**
10. **Case Analysis (case details, outcomes)**
11. **IPC to BNS Mapping (update legal sections)**

Experimental Setup

- **Dataset:** 7100 Indian Supreme Court judgments
- **Models Used:**
  - InLegalBERT (Extractive Summarization)
  - Facebook/BART- (Abstractive Summarization)
- **Evaluation Metrics:** ROUGE, BLEU, METEOR, BERTScore
- **Hardware:** Windows 11, 16GB RAM, Ryzen 6800H, RTX 3050 GPU

Experimental Setup

- **Dataset :**
  - **Data Source:** The dataset was created from 7.1k Indian Supreme Court case documents.
  - **Data Splits:** The data is divided into two splits: 'train' (7k docs) and 'test' (100 docs).
  - **Data Instance:** An example from 'train' looks as follows:


```
{
"id": "2858"
"num_doc_tokens": 4867
"num_summ_tokens": 708
"document": ["Appeal No. 36 of 1967.",
             "Appeal by special leave...", ...]
"summary": ["The appellant executed a usufructuary mortgage ...",
            "In 1949, the mortgagee left for", ...]
}
```


fig. (2) Dataset Instance Example

Experimental Setup

- **Models Used:**
  - InLegalBERT (Extractive Summarization)

➢ This model is used to generate legal sentence embeddings for extractive summary.

➢ The model has the configuration:

- 12 hidden layers
- 768 hidden dimensionality
- 12 attention heads
- ~110M parameters.

https://huggingface.co/law-ai/InLegalBERT

Experimental Setup

- **Models Used:**
  - <u>Facebook/BART- (Abstractive Summarization)</u>
  - ➢ This model is a large variant of BART and is effective for text generation
  - ➢ The model has the configuration:
    - 12 hidden layers in the encoder and decoder
    - 1024 hidden dimensionality
    - 16 attention heads
    - ~406M parameters.

https://huggingface.co/facebook/bart-large

Experimental Setup

- **Evaluation Metrics**
  - ROUGE (Recall-Oriented Understudy for Gisting Evaluation)
  - ➢ ROUGE-1: Measures unigram (single word) overlap between generated and reference summaries.
  - ➢ ROUGE-2: Measures bigram (two consecutive words) overlap.
  - ➢ ROUGE-L: Measures the longest common subsequence of words, reflecting sentence-level structure.

Experimental Setup

- **Evaluation Metrics**
  - BLEU (Bilingual Evaluation Understudy)
  - ➢ Checks how many matching n-grams (phrases) exist between generated and reference summaries.
  - METEOR (Metric for Evaluation of Translation with Explicit ORdering)
  - ➢ Considers synonym matching, word order, and stemming for better fluency evaluation.
  - BERTScore
  - ➢ Uses contextual word embeddings to measure how semantically similar the summaries are.

Output

1. Fine-Tuned BART model evaluation results

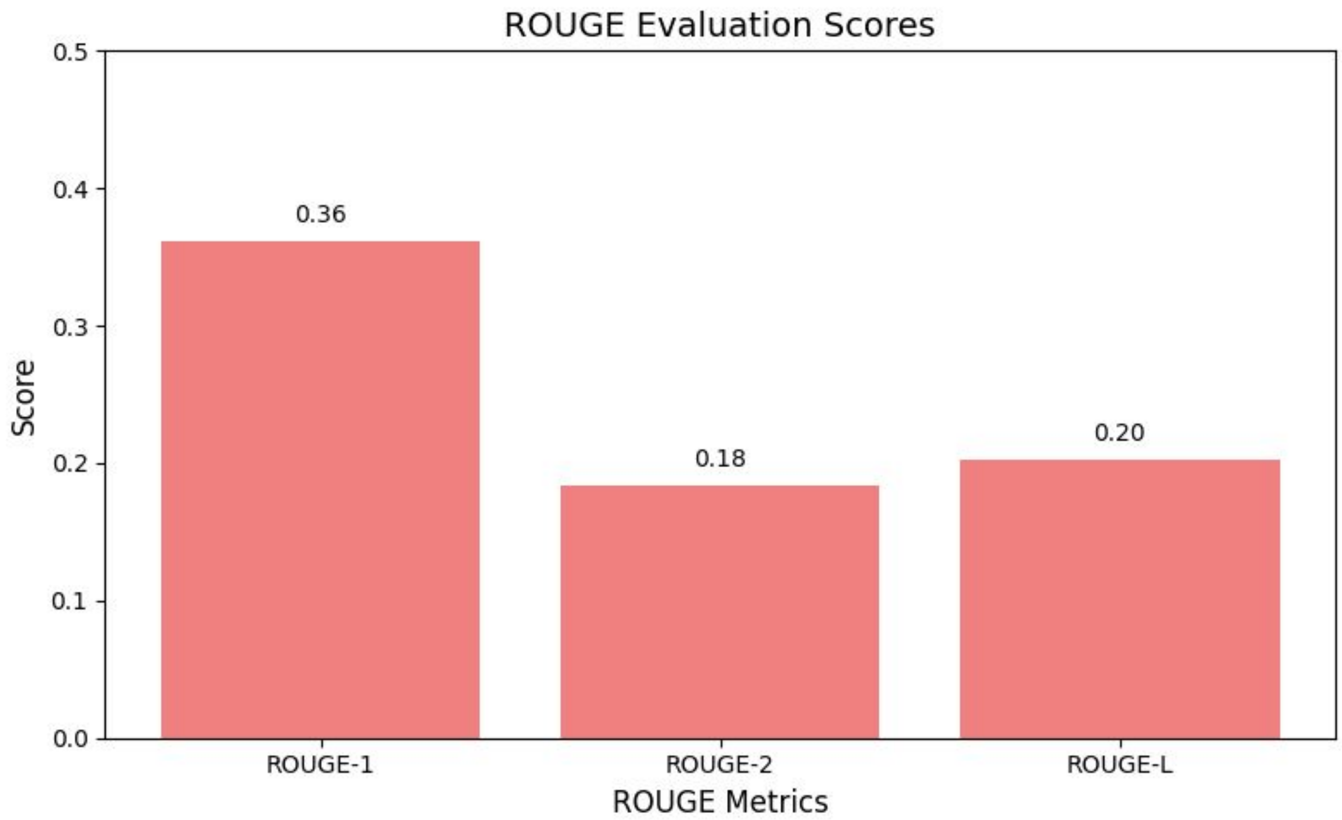


fig. (3) ROUGE evaluations of fine-tuned BART model

Output

1. Fine-Tuned BART model evaluation results

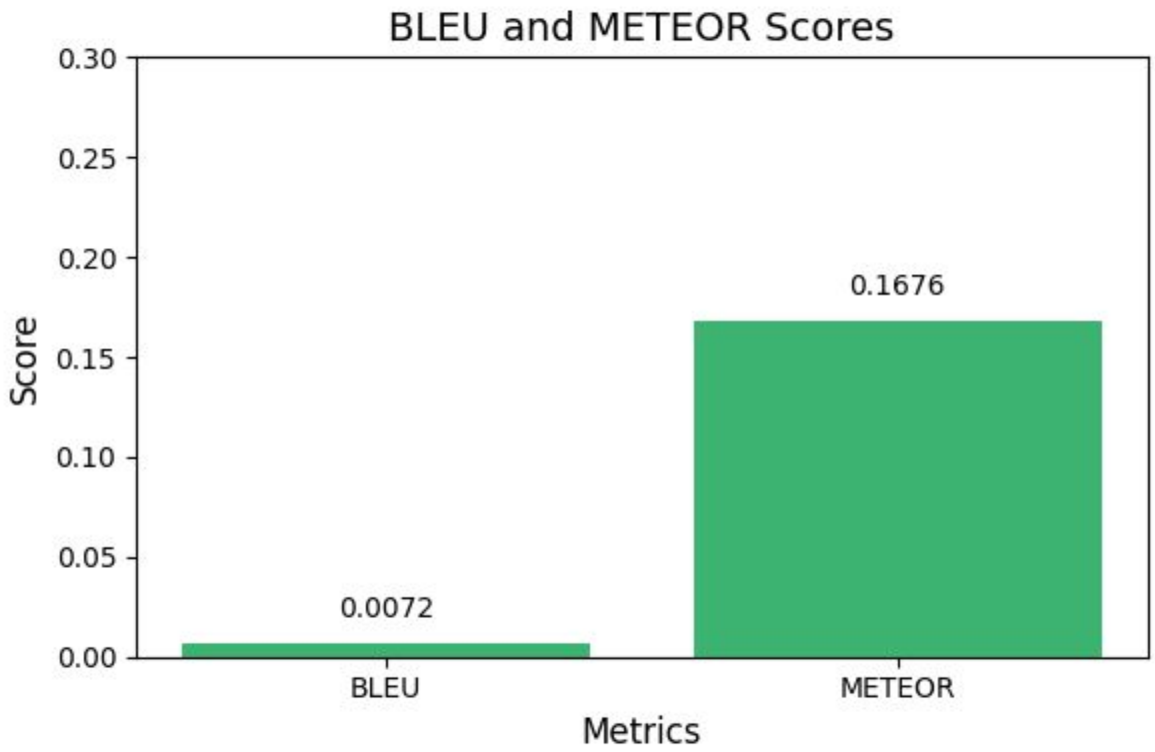


fig. (4) BLEU & METEOR evaluations of fine-tuned BART model

Output

1. Fine-Tuned BART model evaluation results

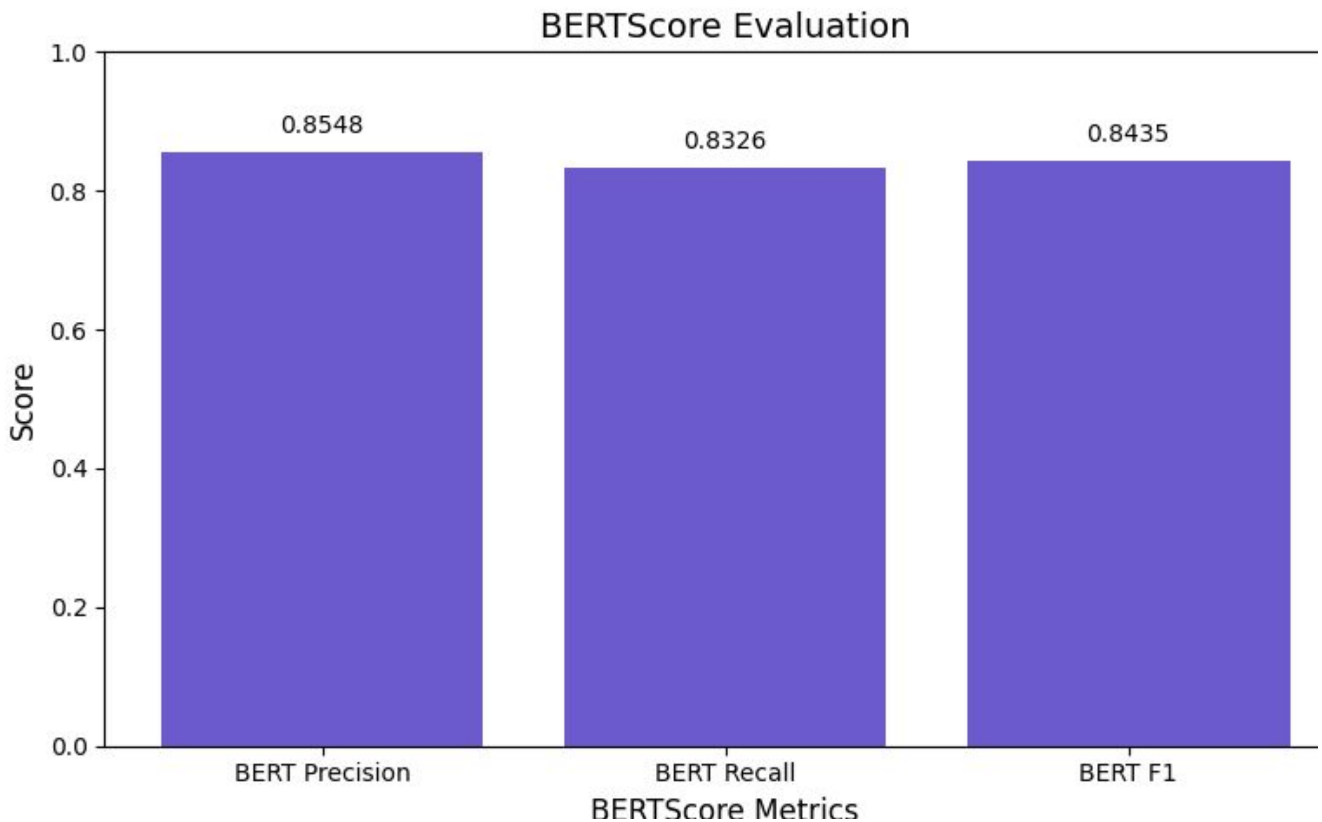
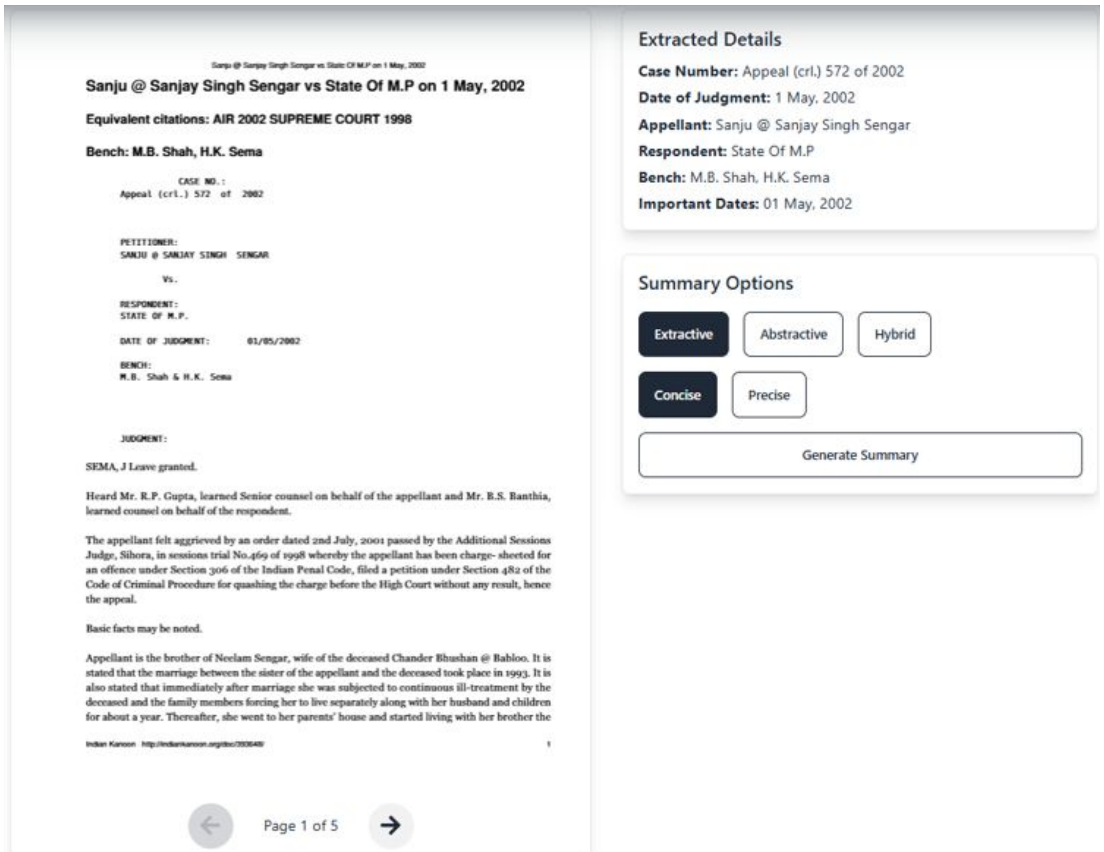


fig. (5) BERTScore evaluations of fine-tuned BART model

Output

2. Case Analysis



User uploads the legal document and gets the case details, as seen in ‘Extracted Details’ section, followed by summary options selection

fig. (6) Document upload and summary option selection

Output

2. Case Analysis

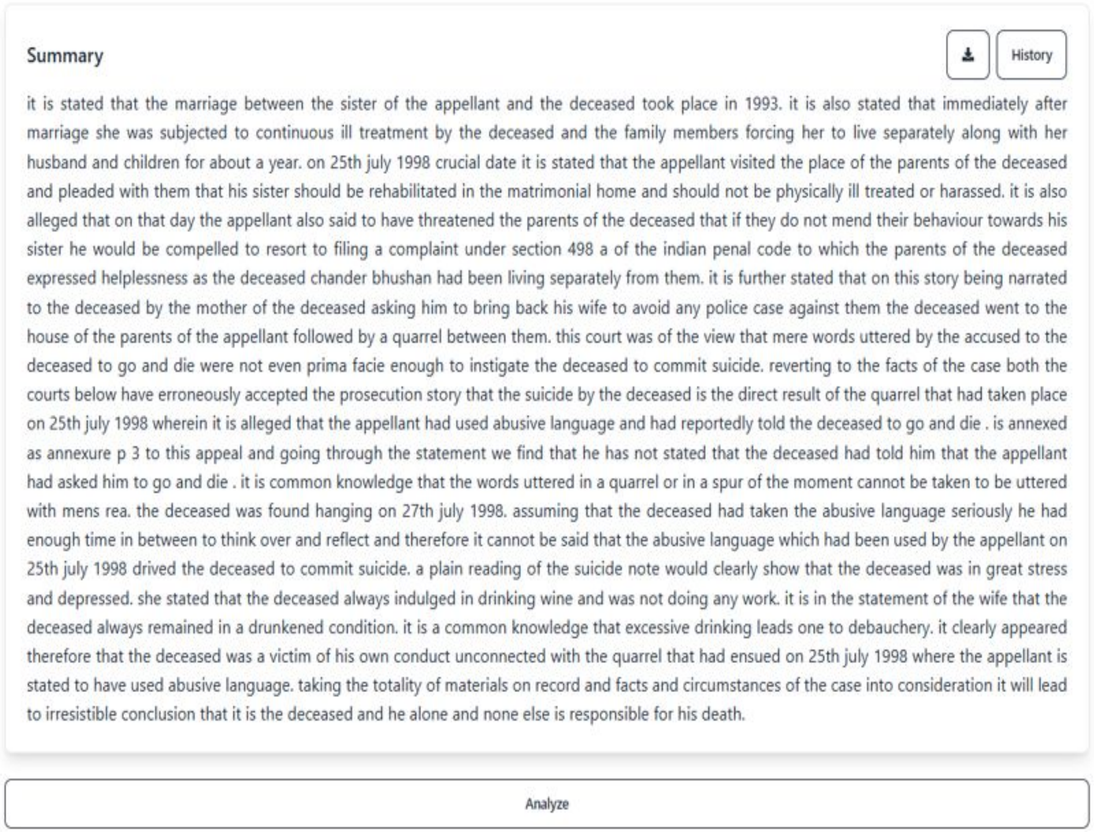


fig. (7) Generated Summary

After selecting, desired summary options, user gets the summary by clicking on 'Generate Summary' button User can get detailed analysis by clicking 'Analyze'

Output

2. Case Analysis

Document Analysis:

Background

The appellant, Sanju Sanjay Singh Sengar, appealed against a charge framed against him for abetting the suicide of his brother-in-law under Section 306 of the Indian Penal Code (IPC). The suicide occurred on July 27, 1998, two days after a quarrel where the appellant allegedly abused and told the deceased to "go and die."

Evidence

The facts involved alleged ill-treatment of the appellant's sister, Neelam Sengar, by her husband (the deceased) and his family. The appellant visited the deceased's parents on July 25, 1998, threatening to file a police complaint under Section 498A of IPC if his sister was not treated well. Following this, a quarrel occurred between the appellant and the deceased where the deceased claimed he was threatened and abused by the appellant. Two days later, the deceased committed suicide and left a suicide note implicating the appellant.

Judgment

Precedent Case

1. **Swamy Prahaladdas v. State of M.P. (1995 Supp. 3 SCC 438)**: Only uttering "go and die" isn't enough to instigate.
2. **Mahendra Singh v. State of M.P. (1995 Supp. 3 SCC 731)**: Harassment allegations alone aren't sufficient under Section 306 IPC.
3. **Ramesh Kumar v. State of Chhattisgarh (2001 9 SCC 618)**: Words spoken in anger without an intent for the consequences should not be seen as instigation.

Important Sections

Case Outcome

fig. (8) Case analysis of document

User gets detailed analysis of the case, like background, evidences, judgment, case outcome, etc.

Conclusion

The project developed an efficient system for legal document summarization and analysis under the Bharatiya Nyaya Sanhita (BNS) framework. By fine-tuning InLegalBERT and BART models, the system ensures accurate section mapping, case analysis, and outcome prediction. It helps legal professionals transition smoothly from IPC to BNS and lays the foundation for future advancements in legal technology.

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1. Paul, S. et al. (2023) “Pre-trained language models for the le-gal domain: A case study on indian law”, Available at: https://arxiv.org/abs/2209.06049.
2. S. Sharma, S. Srivastava, P. Verma, A. Verma, and Sachchida Nand Chaurasia (2023) “A Comprehensive Analysis of Indian Legal Documents Summarization Technique,” https://doi.org/10.1007/s42979-023-01983-y
3. Rani, U. and Bidhan, K. (2021a) “Comparative assessment of extractive summarization: TextRank, TF-IDF and LDA”, Journal of scientific research, doi:10.37398/jsr.2021.650140.

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